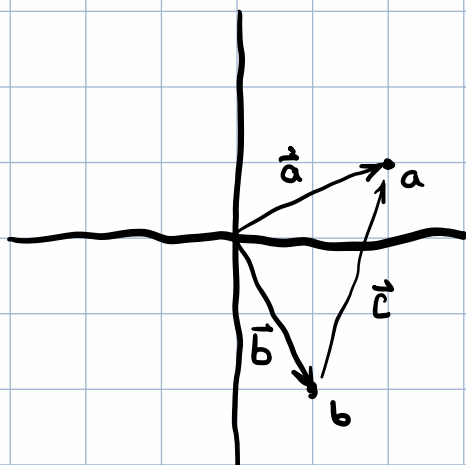


1) Are these vectors orthogonal?

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$a \quad b$

$$a \cdot b = 2 \cdot 1 + 1(-2) = 2 - 2 = 0$$



Check

$|\vec{a}|^2 + |\vec{b}|^2 = |\vec{c}|^2$ iff $\vec{a}, \vec{b}$, and $\vec{c}$ form a right triangle

$$|\vec{a}|^2 = 2^2 + 1^2 = 5$$

$$|\vec{b}|^2 = 1^2 + (-2)^2 = 5$$

$$|\vec{c}|^2 = (2-1)^2 + (1-(-2))^2 = 1^2 + 3^2 = 10 \quad \checkmark$$

2) Find the length of $\vec{a}$.
Normalize $\vec{a}$.

$$\vec{a} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

3-4-5 triangle

$$|\vec{a}| = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$$

$$\vec{a}' = \text{normalize}(\vec{a}) = \frac{\vec{a}}{|\vec{a}|} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

3) Show that $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is normal

$$\left| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right| = \sqrt{1^2 + 1^2} = \sqrt{2} \therefore \frac{\sqrt{2}}{\sqrt{2}} = 1$$

4) Verify that $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ satisfies $Q^T Q = I$.

Show the intermediate matrix multiplication.

$$Q^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad Q^T Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right)$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{2} \left[1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right]$$

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ b \end{bmatrix} + y \begin{bmatrix} c \\ d \end{bmatrix} = \frac{1}{2} \left[\begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right]$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I = Q^T Q$$

5) Given $\vec{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ compute $Q^T \vec{x}$

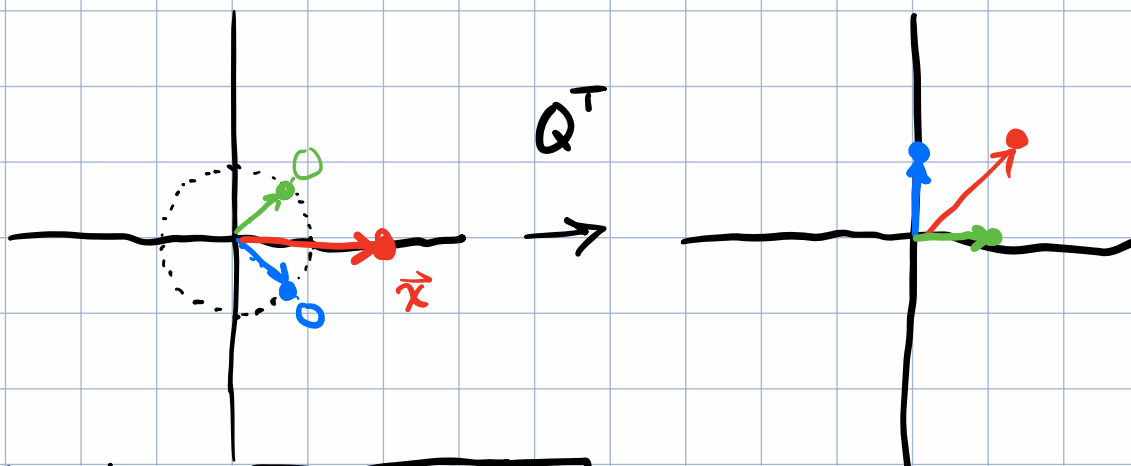
Interpret what the resulting coordinates mean

in terms of the new basis vectors

$$Q^T \vec{x} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} \frac{2}{\sqrt{2}} \\ \frac{2}{\sqrt{2}} \end{bmatrix}$$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



$$|\vec{x}'| = \sqrt{\left(\frac{2}{\sqrt{2}}\right)^2 + \left(\frac{2}{\sqrt{2}}\right)^2} = \sqrt{2+2} = 2$$

$$|\vec{x}| = \sqrt{2^2 + 0^2} = 2$$

$$|\vec{x}'| = |\vec{x}|$$

Q^T results in a rotation $\frac{\pi}{4}$ counter-clockwise plus an inversion of the coordinate system.